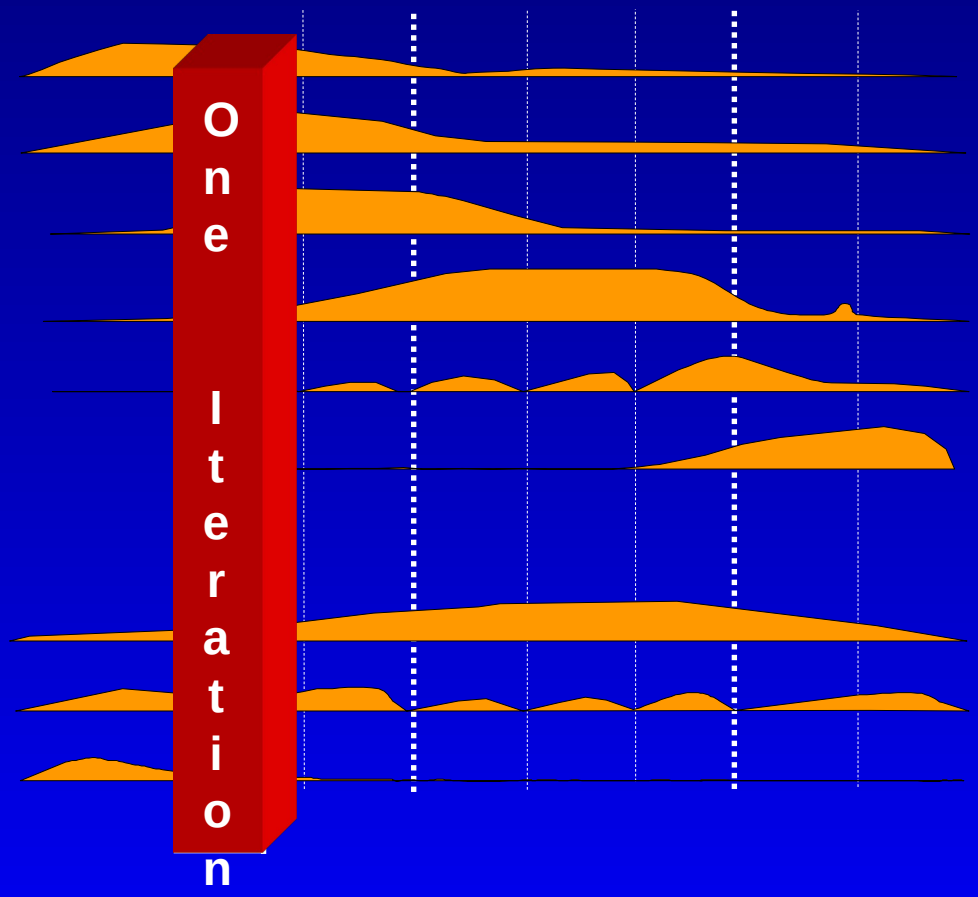
Iteration Workflows



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Objectives: Iteration Workflows

- Describe three typical iteration workflows, one for each of the following phases:
  - Inception (getting started)
  - Elaboration (working on the architecture)
  - Construction (finishing up)
- Understand how the emphasis among workflows changes in different phases
- Understand what an artifact is and distinguish among an artifact, a document, and a report
- Describe the sets of artifacts defined by the Rational Unified Process and how they grow throughout the lifecycle

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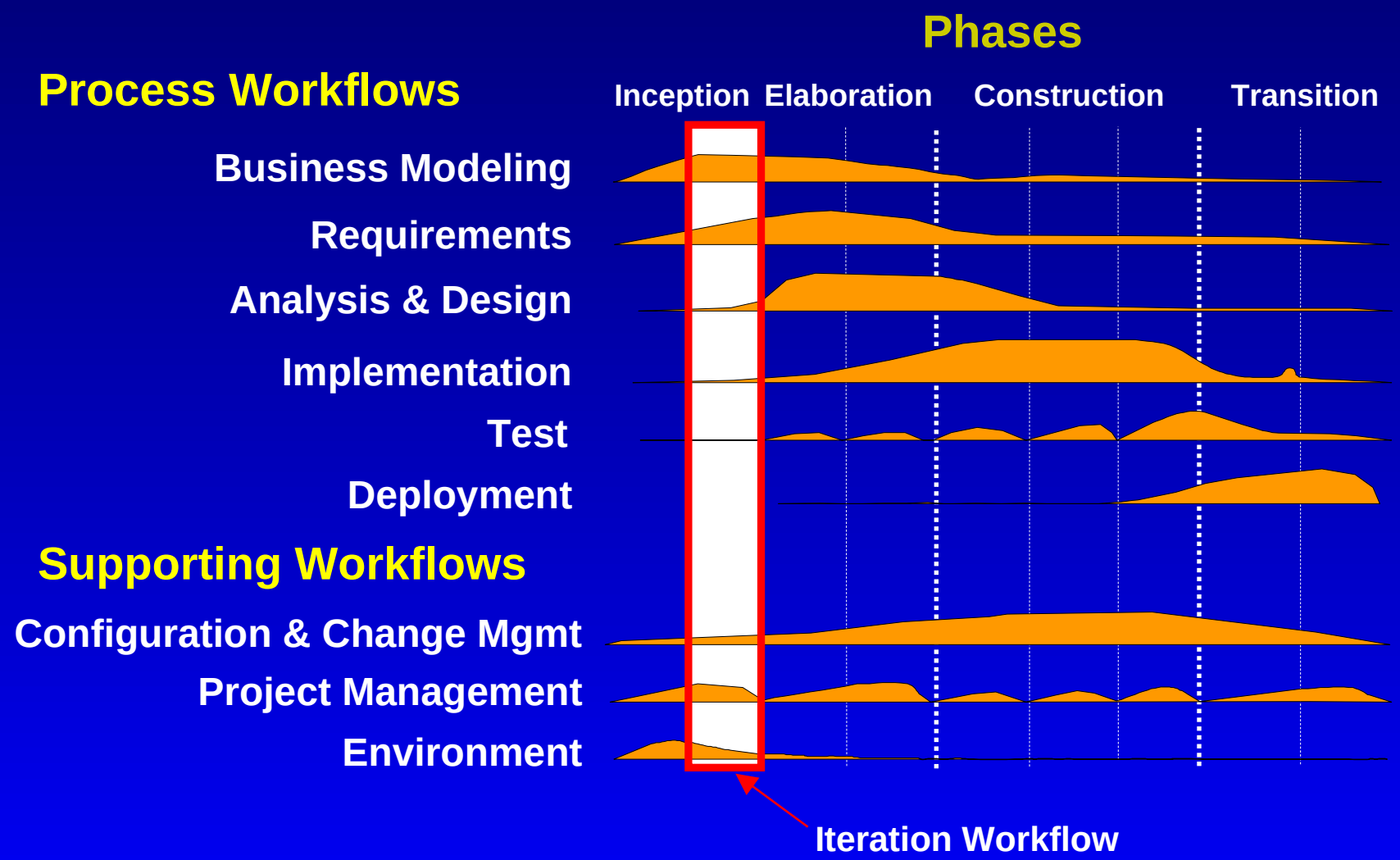
R

A Typical Inception Iteration



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Context and Inception Iteration Objectives

- We are beginning the initial development cycle for a new product
- This is a green field development for a small system. There is no requirement to preserve, reuse, or interface to a legacy system
- The objectives are to define the project vision and the business case
- Most of the effort will be spent in the requirements and project management workflows

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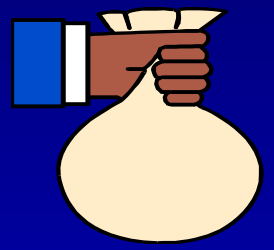
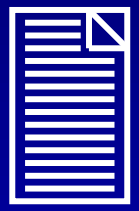
Requirements Activities

Customer

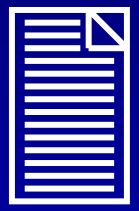


Users

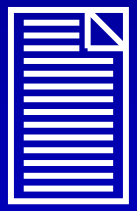
- *Define the vision and scope of the system*
- *Begin the risk list*
- *Collect functional requirements using storyboarding, brainstorming, etc.*



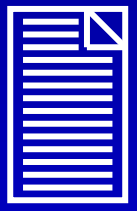
Vision



Business Case

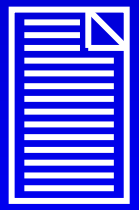


Supplementary Specifications

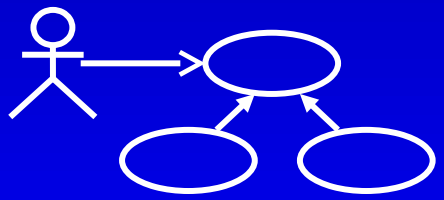


Risk List

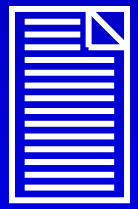
Other stakeholders
Domain Experts
Industry Analysts
Site Visits
Business Model



Stakeholder Needs



Use-Case Model (sketch)



Glossary

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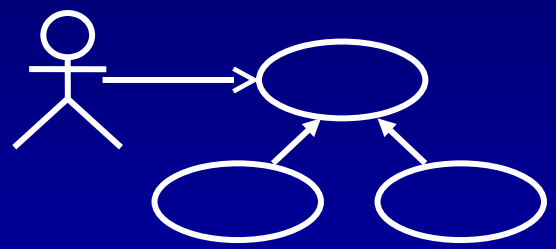
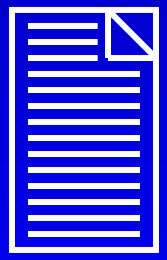
Project Management Activities



Use-Case Model (partial)

Vision

Supplementary Specifications



Risk List

- Consider economic factors: costs, resource estimates, revenue projection, etc.
- List business risks
- Begin planning at the phase level

Business Case

Risk List

Project Plan

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Artifacts of the Inception Phase

- Business Case
- Vision
- Glossary
- Software Development Plan, including at minimum
  - Project Plan
  - Risk List
- Use-Case Model (preliminary)
- Supplementary Specifications (preliminary)
- User Interface Prototype (optional)
- Guidelines, including design, modeling, and programming guidelines
- Development Case

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Evaluating the Inception Iteration

- Since this iteration is the only one planned for inception, the Lifecycle Objective Milestone occurs at the end of the iteration
- Stakeholders review the artifacts produced and a “go” decision is made

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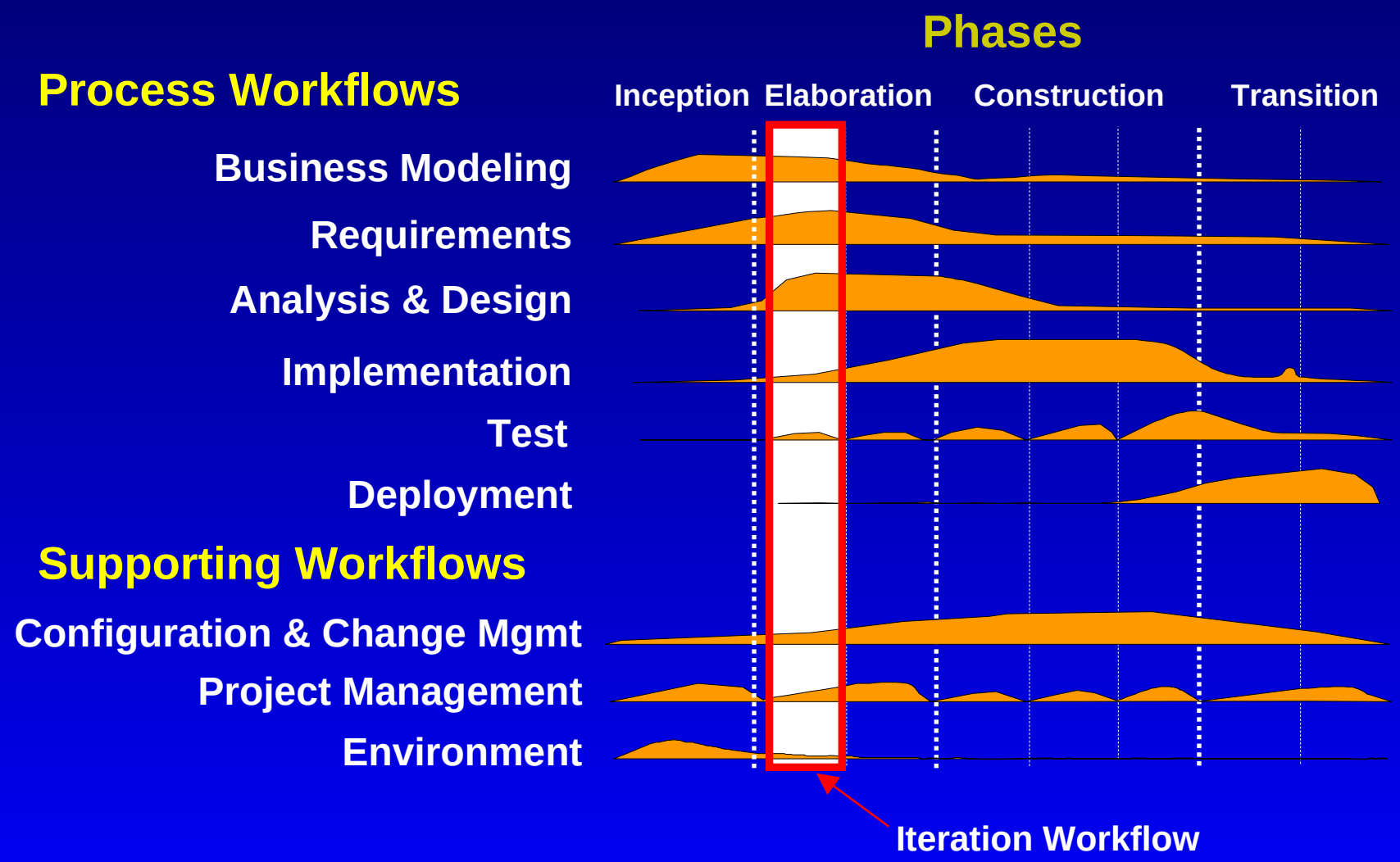
A Typical Elaboration Iteration



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Context and Elaboration Iteration Objectives

- The Lifecycle Objective milestone was passed and a “go” decision was made
- The objectives are to begin to address the architectural risks that have been identified
- Most of the effort will be spent in the requirements, analysis & design, and project management workflows

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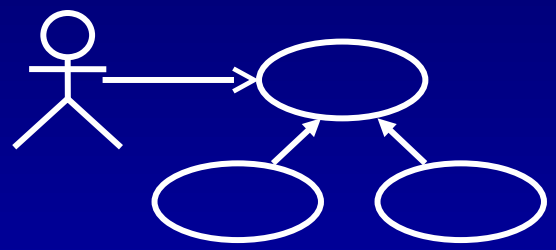
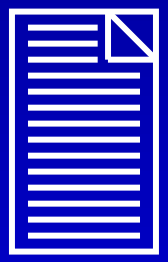
Project Management Activities: Initial



Use-Case Model (partial)



Vision

Project Plan

Risk List

- Project manager prepares an iteration plan including evaluation criteria and a list of use cases that drive the architecture

Iteration Plan

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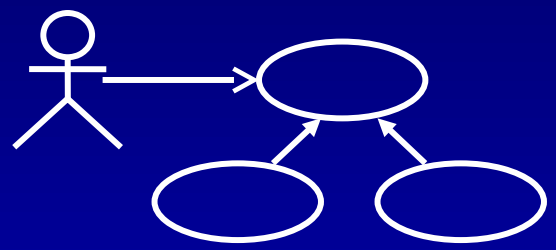
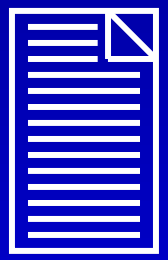
Requirements Activities

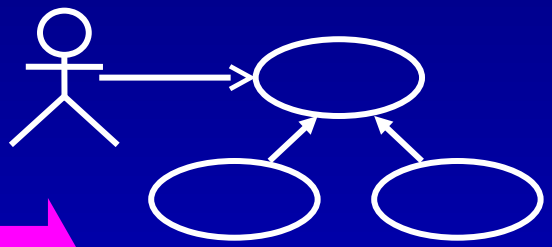


Use-Case Model (partial)

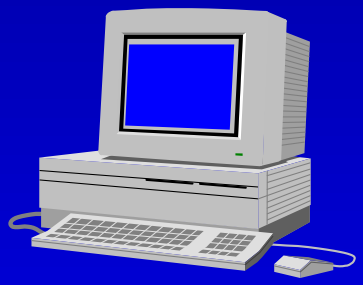


Supplementary Specifications

- Detail the selected use cases
- Begin work on a user-interface prototype
- Continue development of the use-case model



Use-Case Model



User Interface Prototype

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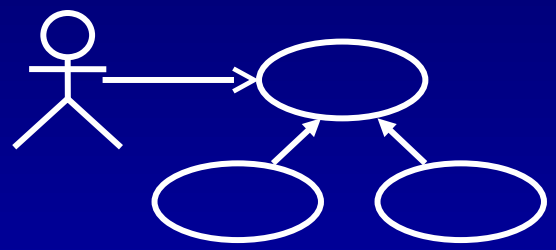
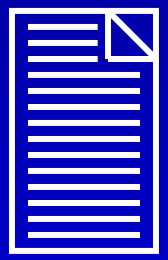
Analysis & Design Activities



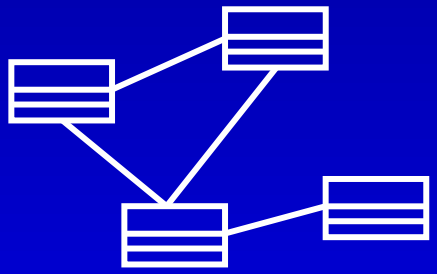
Use-Case Model



Supplementary Specifications

- Identify classes and organize them into design packages
- Define design mechanisms
- Consider concurrency and distribution

Software Architecture Document



Design Model

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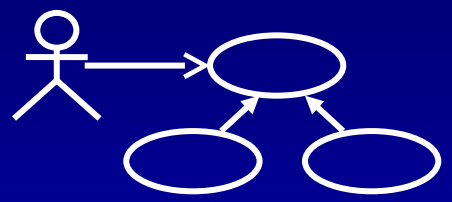
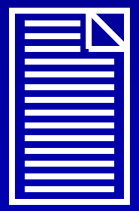
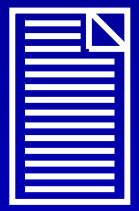
Implementation and Test Activities



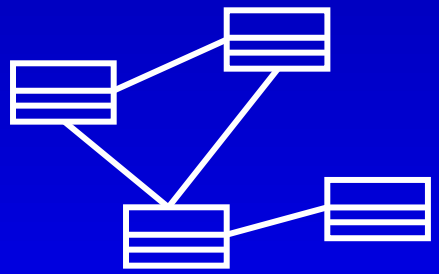
Use-Case Model



Supplementary Specifications

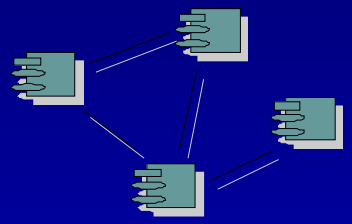


Iteration Plan

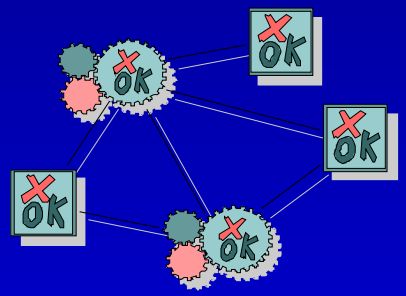


Design Model

- Define the implementation view
- Plan the integration of subsystems
- Plan and develop system and integration tests
- Implement and unit test the classes
- Execute the tests



Implementation Model



Test Model

Integration Build Plan

Test Plan

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Artifacts of the Elaboration Phase

- Business Case (updated)
- Vision (updated)
- Glossary (updated)
- Software Development Plan (updated)
- Use-Case Model (updated)
- Supplementary Specifications
- Software Architecture Document
- Architectural Prototype
- CM Plan
- Analysis Model (optional)
- Design Model
- Data Model (optional)
- Implementation Model
- Test Model
- Change Requests
- For each iteration:
  - Integration Build Plan
  - Test Plan
  - Iteration Plan
  - Iteration Assessment

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Evaluating the Elaboration Iteration

- The architect assesses the results of the tests run relative to the evaluation criteria
- The architect and project manager update the risk list
- The project manager compares the iteration's actual cost, schedule, and content with the plan, determines if any rework will be done, updates the project plan and upcoming iteration plan as needed
- The result is a first cut at the architecture and design lessons learned which will allow the architect to refine the architectural design in subsequent iterations

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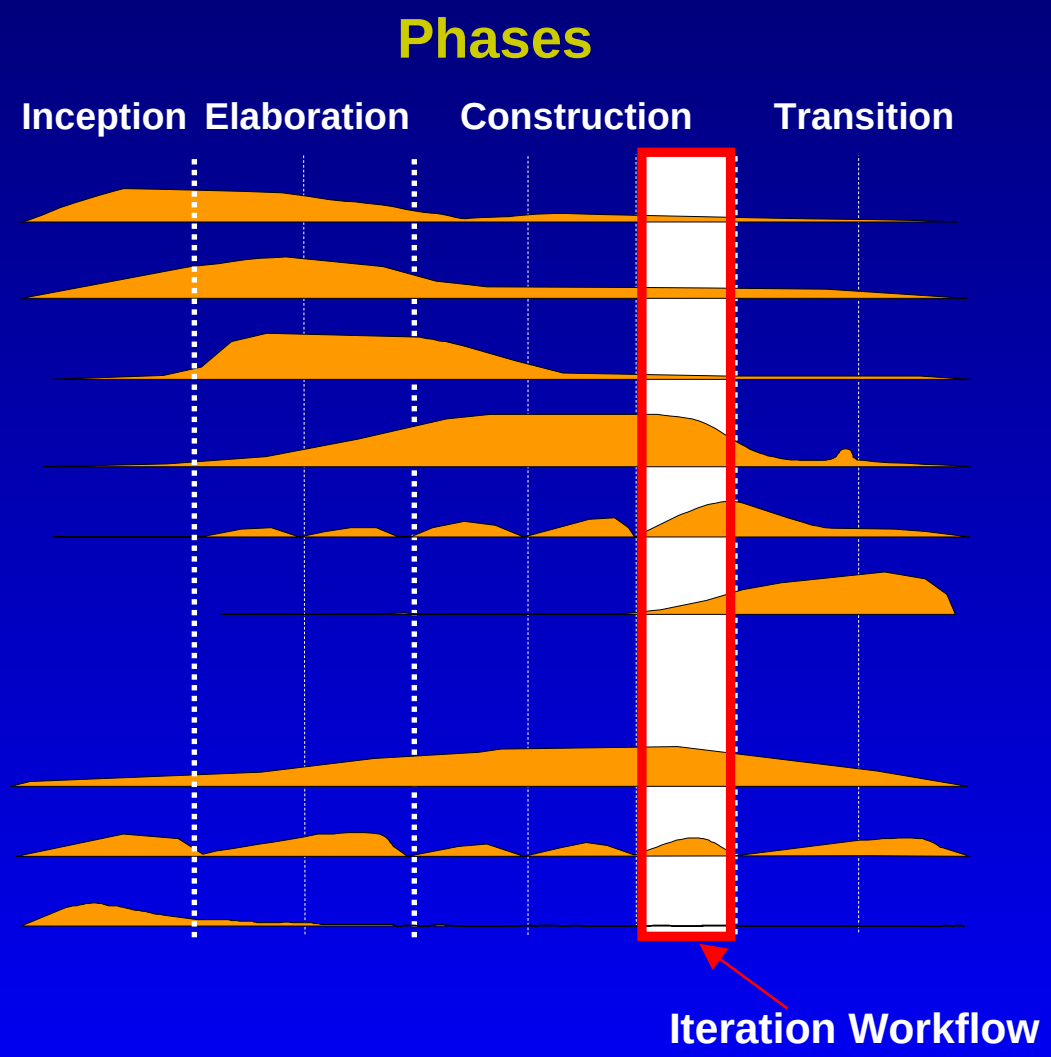
A Typical Construction Iteration



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Context and Construction Iteration Objectives

- It is late in the construction phase. Requirements are stable. The architecture is fully implemented. A great deal of functionality has been implemented and integrated
- The project is approaching its first “beta” release and the IOC milestone
- Most of the effort will be spent in the implementation and test workflows

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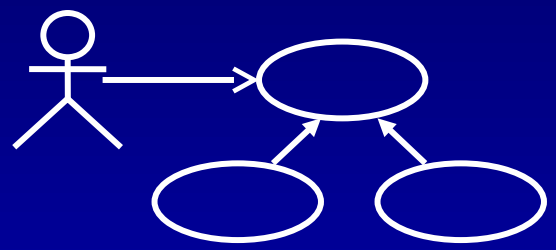
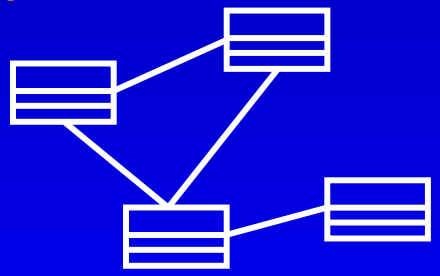
Analysis & Design Activities



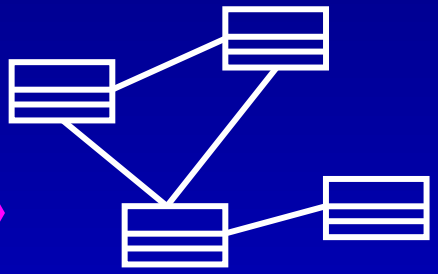
Use-Case Model

Supplementary Specifications



Design Model

- *Detail the remaining use cases*
- *Identify new classes or relationships to be built*
- *Get approval for changes to classes being reused*



Design Model

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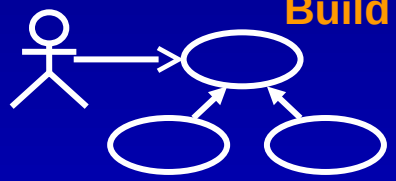
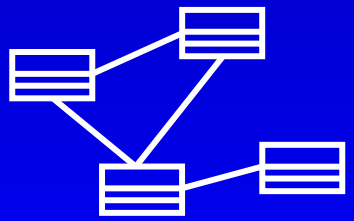
Implementation Activities

Integration Build Plan



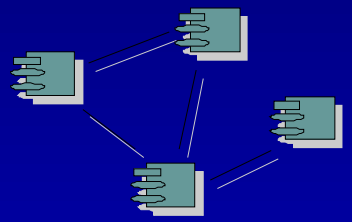
Use-Case Model

Supplementary Specifications

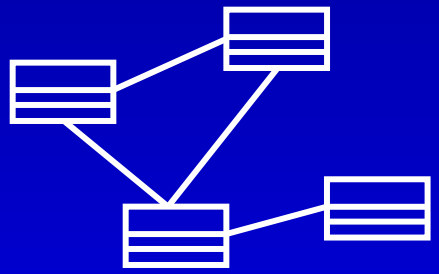


Design Model

- *Implement new classes*
- *Make approved changes to reused classes*
- *Conduct unit tests*



Implementation Model



Design Model

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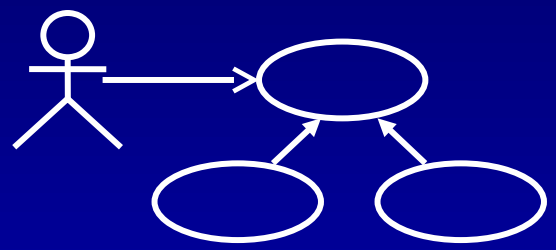
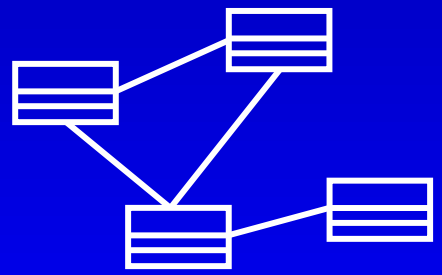
Test Activities



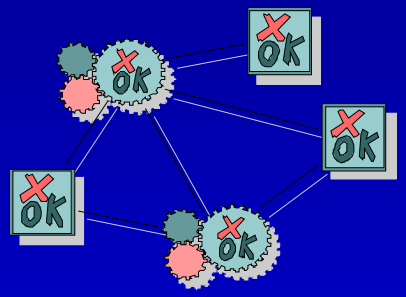
Use-Case Model

Supplementary Specifications



Implementation Model

- Plan and design system and integration tests for new use cases
- Execute the integration and system tests
- Perform regression testing on modified classes



Test Model

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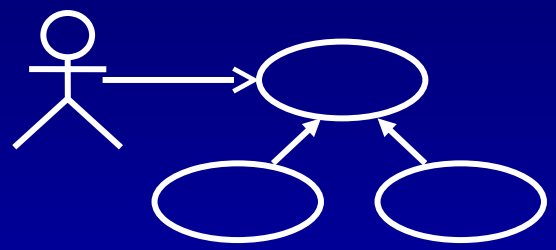
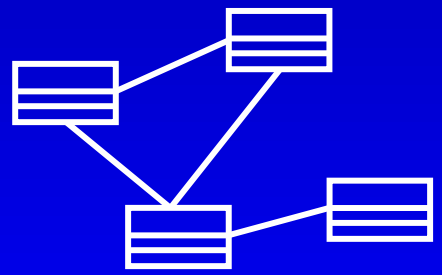
Configuration & Change Management Activities



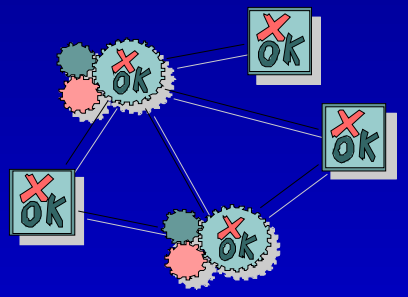
Use-Case Model

Supplementary Specifications



Implementation Model

- *Perform builds*
- *Periodically report status of change requests*
- *Provide timely approval of requested changes*
- *Baseline the product*



Test Model

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Artifacts of the Construction Phase

- Software Development Plan (updated)
- Use-Case Model (updated)
- Software Architecture Document (updated)
- Analysis Model (optional)
- Design Model
- Data Model (optional)
- Implementation Model
- Test Model
- Change Requests
- For each iteration:
  - Integration Build Plan
  - Test Plan
  - Iteration Plan
  - Iteration Assessment
- Deployment Plan
- Training Materials (preliminary)
- End-User Support Material (preliminary)
  - Installation Artifacts
  - Release Notes

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Evaluating the Construction Iteration

- Since this iteration is the last one planned for construction, the Initial Operating Capability milestone occurs at the end of the iteration
- Stakeholders review the progress achieved and a decision is made whether the product is ready for release based on
  - Number of errors remaining
  - Completeness and accuracy of user documentation
  - Readiness of customers to participate in a beta release

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Our Process Is Not Document-Driven

- Many artifacts are produced in each phase and iteration
- Most artifacts of the process are models (e.g., use-case model or design model)
  - Keep them in the appropriate development tool
- Very few artifacts are documents (e.g., plans and status assessments)
- The process also defines reports
  - Reports are documentation generated for review purposes
- Artifacts are baselined and configuration managed; reports are generated when needed and not baselined

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R

Sets: Groupings of Artifacts

- Management set
  - Planning artifacts, such as the business case, software development plan
  - Operational artifacts, such as status assessments
- Requirements set
  - Such as the vision document, use case model, supplementary specifications
- Design set
  - Design model, test model, architecture description
- Implementation set
  - Source code, executables, and associated files needed
- Deployment set
  - All material delivered, such as users manuals, installation procedures, training

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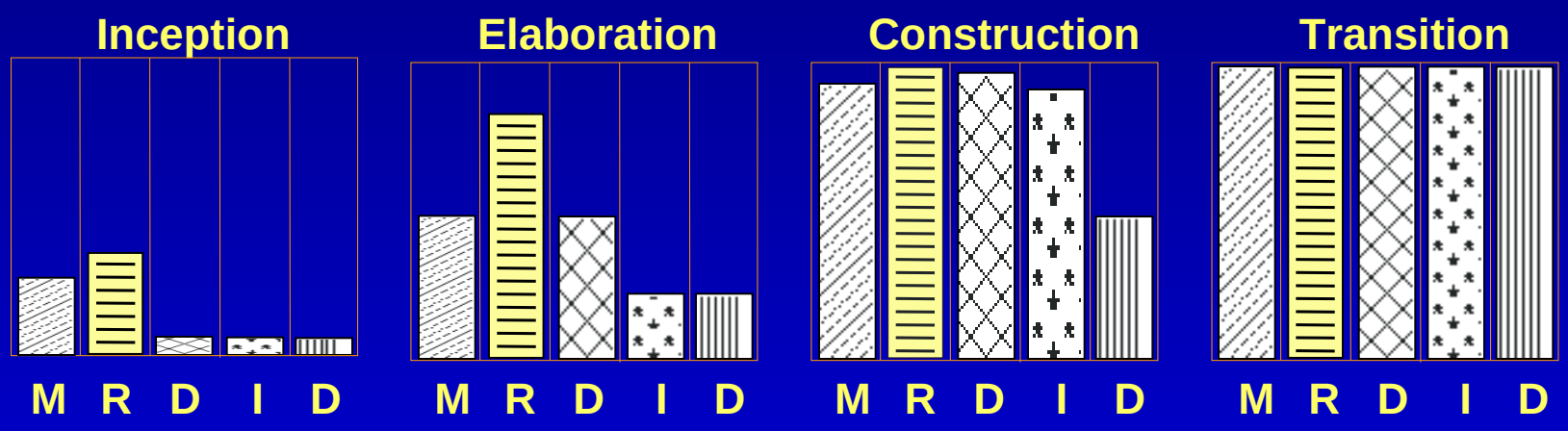
Artifact Sets Are "Grown" throughout the Lifecycle



- M : Management set
- R : Requirements set
- D : Design set
- I : Implementation set
- D : Deployment set

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Summary: Iteration Workflows

- All workflows participate in all iterations
- Some workflows are more important in one iteration than another
- Issues and concerns vary among the iterations as well
- All iterations reflect our six best practices
- The process defines artifacts that evolve over the phases and iterations of the lifecycle
- The process emphasizes models and technical artifacts rather than documents

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